

INCIDENT REPORT 5

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Report ID: IR-005
Incident Name: File Integrity Monitoring Alert — /etc/hosts Modified
Date/Time: May 5, 2026, around 12:17
Severity: MEDIUM
Status: Resolved (lab environment)
Environment: Wazuh 4.7.5 Mini SOC Home Lab
Analyst: Jazz

1. EXECUTIVE SUMMARY

At approximately May 5, 2026 around 12:17, Wazuh SIEM detected a file integrity change on the Ubuntu endpoint. The monitored file /etc/hosts was modified by adding a fake hostname entry.

Wazuh File Integrity Monitoring, also called Syscheck, generated Rule ID 550: "Integrity checksum changed" with severity level 7. This confirmed that Wazuh successfully detected modification of a sensitive Linux system file.

The incident maps to MITRE ATT&CK T1565.001 — Stored Data Manipulation under the Impact tactic.

2. TIMELINE

May 5, 2026 before 12:17

FIM/Syscheck was configured to monitor Linux system directories such as /etc and /usr/bin.

May 5, 2026 around 12:17

The /etc/hosts file was modified on ubuntu-endpoint.

May 5, 2026 around 12:17

The fake entry was added:

192.168.56.1 evil.com

May 5, 2026 shortly after modification

Wazuh generated Rule ID 550: "Integrity checksum changed."

May 5, 2026 after alert review

The alert was opened and reviewed in Wazuh Security Events.

May 5, 2026 after investigation

The event was confirmed as a true positive lab simulation.

3. DETECTION SOURCE

Tool: Wazuh SIEM/XDR

Wazuh Version: 4.7.5

Detection Area: File Integrity Monitoring / Syscheck

Log Source: Wazuh FIM / Syscheck

Affected Agent: ubuntu-endpoint

Agent OS: Ubuntu 22.04.5 LTS

Affected File: /etc/hosts

Observed Wazuh Rule ID:

- 550 — Integrity checksum changed

Severity:

- Rule 550: Level 7

Event Type:

- File modified

Monitored Directories:

- /etc

- /usr/bin

4. MITRE ATT&CK MAPPING

Tactic: Impact

Technique: Data Manipulation

Sub-technique: T1565.001 — Stored Data Manipulation

Explanation:

The activity maps to Stored Data Manipulation because a system file was modified. Attackers may modify stored files or configuration data to redirect traffic, hide activity, block security tools, or alter system behavior.

Potential attacker next steps if malicious:

- Redirect traffic to attacker-controlled infrastructure
- Block access to security update servers
- Modify other system configuration files

- Establish persistence
- Hide malicious activity

5. INVESTIGATION FINDINGS

Source:

Local modification on ubuntu-endpoint.

Target:

The monitored file /etc/hosts.

Change performed:

The following fake entry was added:

192.168.56.1 evil.com

Observed behavior:

- Wazuh detected a checksum change.
- The alert appeared in Security Events.
- The alert was identified as FIM/syscheck activity.
- The affected file path was /etc/hosts.
- Rule ID 550 fired with Level 7 severity.
- The activity mapped to MITRE T1565.001.

Evidence reviewed:

- Wazuh FIM alert table
- Expanded FIM alert details
- File path /etc/hosts
- Syscheck fields
- MITRE ATT&CK Stored Data Manipulation dashboard

Compromise confirmed:

No. The file change was intentionally generated in the lab.

Unauthorized modification confirmed:

No production unauthorized activity occurred.

6. RISK ASSESSMENT

Risk: MEDIUM

Reason:

Modification of `/etc/hosts` is suspicious because it can affect hostname resolution and redirect traffic. In production, this could indicate tampering, malware activity, persistence, or attempted disruption.

The risk is higher if:

- The change was unauthorized
- The file redirects security domains
- Other sensitive files were modified
- The change followed sudo privilege escalation
- The modified file persists after reboot

In this lab, the risk was controlled because the change was intentional.

7. VERDICT

Verdict: TRUE POSITIVE — Lab Simulation

The alert is a true positive because `/etc/hosts` was actually modified and Wazuh correctly detected the file integrity change. It was not a false positive because the checksum changed after the file was edited.

8. RESPONSE ACTIONS TAKEN

1. Reviewed Wazuh Security Events.
2. Filtered for FIM/syscheck activity.
3. Identified Rule ID 550.
4. Verified the affected agent as ubuntu-endpoint.
5. Verified the affected file as `/etc/hosts`.
6. Reviewed expanded FIM alert details.
7. Confirmed MITRE mapping to T1565.001.
8. Documented the incident.

9. RECOMMENDATIONS

1. Monitor critical files using FIM:
 - /etc/hosts
 - /etc/passwd
 - /etc/shadow
 - /etc/sudoers
 - /etc/ssh/sshd_config
2. Investigate unauthorized changes to /etc/hosts.
3. Compare modified files against a known-good baseline.
4. Review sudo logs to identify who modified the file.
5. Restore unauthorized changes.
6. Alert on critical file modifications.
7. Correlate FIM alerts with sudo activity.
8. Restrict write access to sensitive files.
9. Monitor for multiple file changes in a short time.
10. Include FIM alerts in daily SOC review.

10. FINAL ANALYST NOTE

This incident demonstrates Wazuh's File Integrity Monitoring capability. Wazuh successfully detected modification of a sensitive Linux system file, generated an alert, and mapped the activity to MITRE ATT&CK. This proves the Mini SOC lab can detect file tampering.